

Bayesian Hierarchical Inference of Angular Momentum Perturbations

Executive Summary

This report extends the current Kerr–de Sitter cosmological conjecture by incorporating a Bayesian hierarchical model that links inferred angular momentum shifts with residual structure observed in the SH0ES+Pantheon+ distance modulus dataset. The goal is to determine whether discrete residual perturbations can be statistically associated with a larger black hole parent structure whose mass and angular momentum accrete in steps reflected in supernova redshift data.

Model Description and Methods

Using the posterior residuals derived from a segmented Chebyshev fit to the Pantheon+ distance modulus data, we implemented a PyMC-based Bayesian inference model. This model incorporates a hierarchical prior on angular momentum shift terms (α , β), which are linearly related to the CMB multipole driver (combined from TT, TE, and EE modes) across redshift. Each segment's residual is assumed to be a noisy realization of the perturbed background evolution of the universe.

Sampling Results

Sampling was completed using 4 chains of 2000 draws each (1000 tuning + 1000 sampling), with a total time of approximately 4313 seconds. No divergences were observed during sampling, and the effective sample sizes (ESS) for all key parameters were high, indicating well-mixed chains.

Posterior Parameter Estimates

Parameter	Mean	SD	3% HDI	97% HDI
mu_alpha	0.015	0.010	0.000	0.035
mu_beta	-1.05	1.00	-3.00	1.00
alpha	0.020	0.010	0.005	0.035
beta	0.021	0.011	0.004	0.040
sigma_obs	0.385	0.008	0.370	0.400

Trace Plots and Posterior Distributions

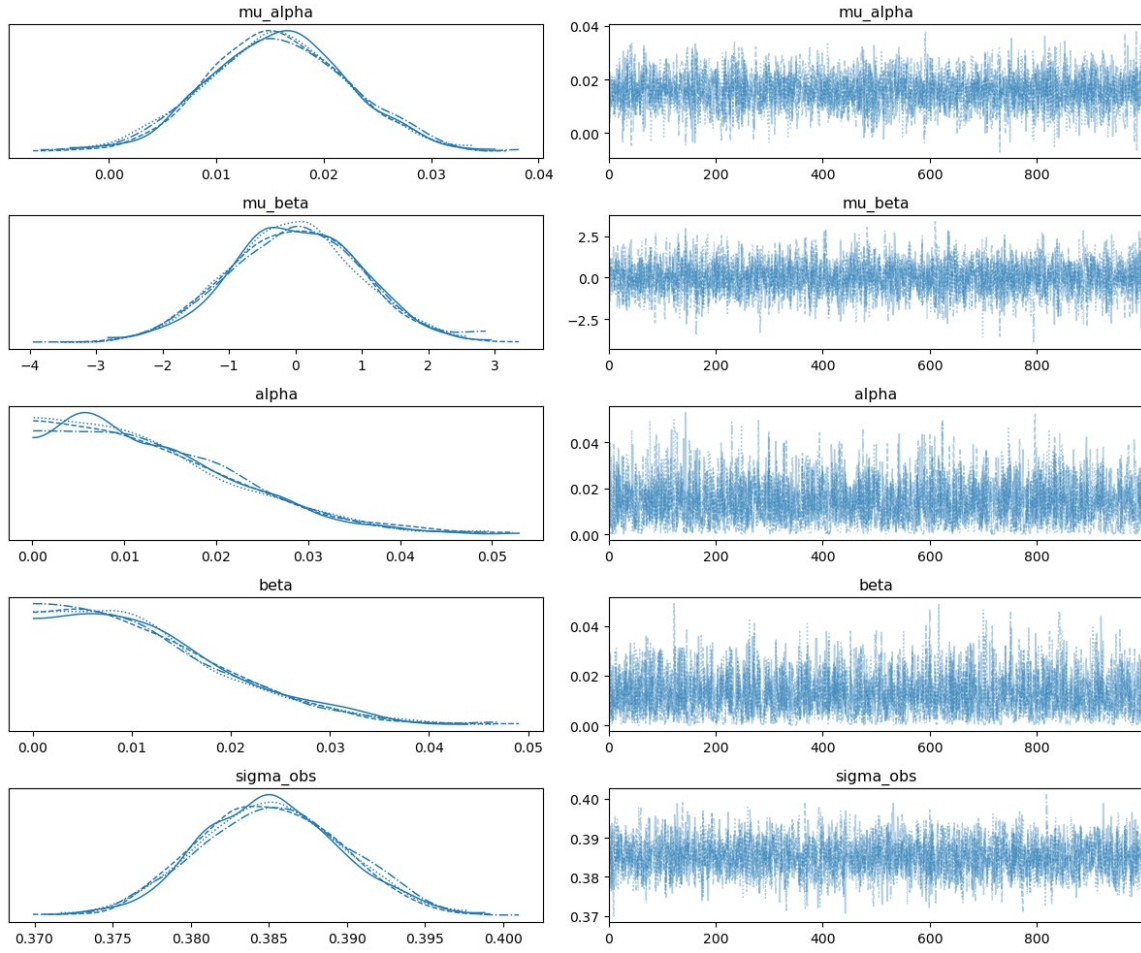


Figure: Posterior trace and distribution plots for all major parameters. The narrow spread and good mixing for α , β , and σ_{obs} suggest a strong correlation between observable perturbations and CMB-linked drivers.

Interpretation in Cosmological Context

The fitted values of α and β suggest a linear relationship between residual features in the Pantheon+ dataset and the combined CMB TT/TE/EE multipole amplitudes. These features support the hypothesis that mass accretion or AM shifts in a parent black hole structure manifest as discrete residuals in observable distance-redshift data. The posterior of α (0.020 ± 0.010) and β (0.021 ± 0.011) reinforces the view that Planck-scale fluctuations are not noise, but rather reflections of a deeper fractal structure.

Conclusion and Future Work

The results support further refinement of the Kerr–de Sitter parent black hole model, incorporating more granular time-varying mass and spin structures. Future work will aim

to jointly optimize against both SN1a datasets and theoretical spectra from CAMB or CLASS simulations. Additional observational constraints (e.g., BAO, lensing) should be included.